Contactar

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www.linkedin.com/in/dev-ai-erick (LinkedIn)
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Aptitudes principales

AI Software Development

Quantitative Research

Writing

Certifications

CCNA Routing and Switching: Introducción a redes

AZ-900: Microsoft Azure Fundamentals

Honors-Awards

NVIDIA Llama Index Contest

Publications

Comparison of artificial vision algorithms in the classification of skin diseases using neural networks

Generative AI and Transformers in Advanced Skin Lesion Classification applied on a mobile device

Skin Disease Pre-diagnosis with Novel Visual Transformers

Erick Garcia

LLM Engineer · Ph.D. Researcher · Building Intelligent Systems with Transformers, ViT, and LangChain
México

Extracto

I'm Erick García, a Ph.D. candidate in Computer Science and remote LLM & AI Solutions Engineer focused on building creative, high-impact AI systems.

My work sits at the intersection of deep learning, backend development, and generative applications. I specialize in:

Transformers & LLMs (Flan-T5, Mistral, custom LoRAs)

Vision models (ViT, GANs, OpenCV, Stable Diffusion)

Tabular + sequential modeling (TFT, LightGBM)

LangChain, Hugging Face, Python, Django, PostgreSQL

Deployment tools: Hugging Face Spaces, Gradio, Streamlit

Experiencia

Su-IT Software a la Medida
Senior AI / LLM Engineer
mayo de 2026 - Present (6 meses)

Secretaría de Ciencia, Humanidades, Tecnología e Innovación Secihti
Doctoral Researcher – PhD in Computer Science
febrero de 2025 - Present (1 año 9 meses)
México

Tech Stack

Conducting PhD research in intelligent agriculture, greenhouse monitoring, and data-driven irrigation analysis.

Designed and deployed a Raspberry Pi-based sensing platform for tomato greenhouse monitoring in Texcoco, Mexico, collecting soil moisture, temperature, humidity, radiation, and root-zone data at 5-minute intervals.

Developed a reproducible Python pipeline for greenhouse time-series analysis, including data cleaning, anti-leakage chronological splits, YAML-based experiment configuration, feature engineering, model training, evaluation, and reporting.

Applied K-means clustering to 14,317 real-world sensor records and identified four interpretable soil moisture regimes. Detected 82 inferred irrigation events and found that 62.5% occurred after deficit-compatible conditions had already appeared, suggesting reactive irrigation timing.

Extended the research toward predictive modeling and decision-support systems by benchmarking XGBoost, LightGBM, GRU, TCN, Transformer, and Temporal Fusion Transformer models for agronomic alerts and greenhouse event forecasting.

Designed and evaluated modified model variants, including custom Regalia-GBDT architectures and adapted sequence models, to compare feature-engineered tabular approaches against deep learning methods for greenhouse time-series prediction.

Integrated Open-Meteo and NASA POWER climate data to enrich greenhouse observations with external weather, radiation, precipitation, wind, vapor pressure deficit, and derived agronomic indicators.

Su-IT Software a la Medida
AI / LLM Engineer
mayo de 2025 - mayo de 2026 (1 año 1 mes)
Ciudad de México, México

Designed, built, and deployed TPSM, a custom Django platform for transfer pricing study management, document automation, report editing, and AI-assisted analysis.

The system centralizes client data, studies, annexes, declarations, report text structures, document imports, spell-check workflows, and HTML/PDF/Word report generation. I also developed a local RAG/AI module to index internal

and external sources, retrieve relevant context, and generate formal Spanish-language economic, industry, and sector analysis texts.

Tech stack included Django 4.2, PostgreSQL, Django REST Framework, LangChain, ChromaDB, sentence-transformers, PyTorch, Transformers, LLaMA 3, Mistral 7B, Flan-T5, PyMuPDF, python-docx, WeasyPrint, Playwright, Pandas, Vanilla JavaScript, Apache, Waitress, and Windows Server.

Key results:
- Reduced tax analysis report preparation from around 1 week to approximately 40 minutes.
- Reduced final client report generation from around 3 days to approximately 20 minutes.
- Automated document ingestion, text extraction, report assembly, formatting, validations, and quality review workflows.
- Built a custom interactive report editor with dynamic sections, modals, editable tables, annex handling, inline validation, and versioned text structures.
- Deployed a CPU-only local AI workflow suitable for a Windows Server production environment.

Outlier
AI Training for Coding
febrero de 2025 - mayo de 2025 (4 meses)
Estados Unidos

Contributed to the training and improvement of generative AI models by evaluating code-related outputs. Responsibilities included:

-Reviewing and analyzing Python code generated by large language models (LLMs).

-Verifying code logic, with a particular focus on conditional structures such as if statements.

-Assessing the reliability, accuracy, and overall quality of code solutions.

-Identifying errors and inconsistencies in the generated code.

-Reporting detected issues and rewriting faulty code to provide correct and optimized solutions.

Edutec Sistemas y Redes

Software Developer

marzo de 2024 - marzo de 2025 (1 año 1 mes)

CALLE BEGONIA 119 SN50010 Toluca de Lerdo, Mexico

TKey Achievements

Contributed to the development of three core modules (Own Revenues, Inventory Management, Asset Control) for a UAEM management system, improving administrative efficiency.

Enhanced data processing and workflow automation using PHP, JavaScript, and CodeIgniter, reducing manual tasks and optimizing system performance.

Developed and optimized database queries and system functionalities, significantly improving data retrieval speed and system responsiveness.

Provided technical support and personalized training to up to 10 users, ensuring effective adoption and use of the system.

Collaborated with cross-functional teams to gather requirements and translate them into technical specifications aligned with institutional processes.

Universidad Autónoma del Estado de México

Full Stack Developer

septiembre de 2023 - agosto de 2024 (1 año)

Instituto Literario 100, Centro, 50000 Toluca de Lerdo, Méx.

Developed a full-stack solution for an interactive socio-economic map of the State of Mexico, showcasing poverty levels, education rates, and marginalization metrics.

Built dynamic, user-friendly front-end interfaces with Vue.js and Vuetify.

Designed and implemented backend services with Node.js and Express.js, ensuring seamless data processing and efficient API integration.

Optimized performance and scalability to handle large datasets, programming the entire solution in JavaScript.

Secretaría de Ciencia, Humanidades, Tecnología e Innovación Secihti

AI Research Fellow – Master’s in Computer Science (Conacyt Scholar)
enero de 2022 - julio de 2024 (2 años 7 meses)
México

Key Achievements

Conducted applied research in computer vision, natural language processing, and generative AI models as part of a Master’s in Computer Science funded by Conacyt.

D.A.N.N.: Built a skin disease detection system using Vision Transformers (ViT) enhanced with GANs and Stable Diffusion, achieving 92.7% validation accuracy and resulting in 3 peer-reviewed publications.

Abracadabra: Developed a Retrieval-Augmented Generation system with a 900K-song dataset for dynamic playlist generation, ranking in the top 200 of the NVIDIA LLaMA Index Contest.

Color Shade: Created a real-time color palette recommendation engine for clothing analysis using ViT, OpenCV, and LightGBM, enabling personalization in fashion applications.

Educación

Universidad Autónoma del Estado de México

Ph.D. in Computer Science, Computer Science · (febrero de 2025 - febrero de 2028)

Universidad Autónoma del Estado de México

Master's degree, Computer Science · (enero de 2022 - julio de 2024)

Tecnologico de Estudios Superiores de Chimalhuacan

Computer Systems Engineer · (junio de 2016 - agosto de 2020)